ModelIt!

Student Activity Pack

AI, Robotics & Ethics — Modeling Hard Choices

When Technology Meets Nature, Who Decides What's Right?

Grade NE | NGSS HS-ETS1-1, HS-ETS1-3, HS-LS2-7, HS-LS4-6 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In Level 1, you modeled whether intervention is justified using severity, risk, and evidence. What NEW factors matter when the technology making decisions is AI — not a human?**

2. **Should an AI system be allowed to make life-or-death decisions about wildlife without human approval? Why or why not?**

3. **What does 'transparency' mean when it comes to technology? Why might people demand it?**

4. **Can you think of a technology that was created to help but ended up causing harm? What went wrong?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Technology Capability Level | Transparency Level | Ecological Crisis Severity | Regulation Strictness | Ecological Benefit | Unintended Consequences | Animal Autonomy Impact | Public Trust | Ethical Justification Score

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Transparent and Regulated

Moderate tech, open methods, strict rules

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Unregulated Cowboy

High tech, low transparency, no rules

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Emergency Override

Maximum tech, extreme crisis, relaxed rules

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Community Benefit — Whether local and Indigenous communities benefit from or are harmed by the technology deployment

Data Security — How well the technology protects sensitive ecological data from misuse, poaching, or commercial exploitation

Long-Term Dependency — Whether ecosystems become reliant on the technology to survive, reducing natural resilience

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When AI and robots are used to protect wildlife, who decides what is ethical — the engineers, the communities, the government, or the algorithms themselves? Use evidence from your simulation to support your answers.

1. **How is the Level 2 model different from the Level 1 ethics model? What does adding 5 more components reveal about the complexity of real ethical decisions?**

2. **Trace the feedback loop: Unintended Consequences → Public Trust → Regulation → Ecological Benefit. What makes this loop so difficult to manage?**

3. **In the Emergency Override scenario, the ethical justification shifts. Is it ever acceptable to lower ethical standards during a crisis? Where do you draw the line?**

4. **How does animal autonomy factor into ethical justification — and why do some people weight it more heavily than others?**

5. **If you were advising a government on AI conservation policy, what would your top three recommendations be based on your model results?**

8 STEM Challenge: Speculative Design Project

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YOUR ENGINEERING MISSION

Design a fictional conservation technology of the future, then evaluate it through your ethical model. Produce a technical blueprint, ethical impact assessment, public communication plan, and a 2-minute pitch video defending your design.

THE SCENARIO

A global conservation council is accepting proposals for next-generation AI-powered conservation technologies. Your team must design a system that addresses a real ecological crisis, but you must also prove that your technology is ethically justifiable. The council will evaluate your proposal on ecological benefit, ethical safeguards, and public communication quality.

Engineering Design Thinking

1. WHAT LEVEL OF AUTONOMY DOES YOUR TECHNOLOGY HAVE — DOES IT RECOMMEND ACTIONS OR TAKE THEM INDEPENDENTLY?

2. HOW DOES YOUR DESIGN PROTECT ANIMAL AUTONOMY WHILE STILL ACHIEVING ECOLOGICAL BENEFIT?

3. WHAT REGULATORY FRAMEWORK WOULD YOU PROPOSE FOR GOVERNING YOUR TECHNOLOGY?

4. HOW WOULD YOU COMMUNICATE YOUR TECHNOLOGY'S PURPOSE AND RISKS TO A SKEPTICAL PUBLIC?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

AI Ethics Researchers and Conservation Technology Policy Analysts develop frameworks for responsible deployment of AI and robotics in environmental contexts. They work with tech companies, government agencies, and NGOs to ensure emerging technologies protect both ecosystems and civil liberties. They earn $85,000–$150,000/year.